

Amirjon Abdunayimov

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EDUCATION:

New York University — *Tandon School of Engineering*

Expected Graduation: May 2028

Bachelor of Science in Computer Science (Minor in Mathematics)

Relevant Coursework: Data Structures & Algorithms, Linear Algebra, Probability & Statistics, Discrete Mathematics

Academic Honors: Jack Kent Cooke Transfer Scholarship Semifinalist, Honors Scholar Designation, Phi Theta Kappa, Dean's List.

WORK EXPERIENCE:

Boardwalk Labs — *Software Engineering Intern*

June 2026 – Present

- **Led the frontend architecture for OctoPilot AI**, building a reusable UI engine across 20+ core Flutter components to eliminate 15+ cross-component regressions.
- **Designed and deployed LLM pipelines and automated text-extraction modules in Python**, establishing semantic knowledge graphs that cut manual processing times by 40%.
- Spearheaded Git-based integration standards across a 5-engineer team, streamlining production-ready releases across **10+ AI learning modules**.
- **Optimized cross-platform application state by implementing a persistent theme engine**, driving visual and interface consistency across the entire user environment.

Sound of Earth — *DevOps & Cloud Infrastructure Intern*

May 2026 – Present

- Architected containerized CI/CD pipelines using Docker and GitHub Actions on Linux, automating multi-environment deployments for **\$15K in annual cloud infrastructure assets**, cutting production release regressions by 40% and saving **\$3,200/year in staging overhead**.
- Developed distributed Python and JavaScript telemetry dashboards to monitor live microservices scaling to **12K+ active platform users**, reducing critical incident detection times by 35% and boosting development velocity by 25%.

Kingsborough Community College Learning Center — *Computer Science & Mathematics Tutor*

March 2026 – June 2026

- Mentored **50+ students** in **algorithmic reasoning**, **data structures**, and debugging paradigms over **100+ sessions** to improve system implementation velocity.
- Designed step-by-step algorithmic exercises covering **matrix operations**, **probability distributions**, and **recursion** to translate mathematical theories into optimized, executable code.

PROJECTS:

StateRelay — **Cross-platform distributed workspace state synchronization system** | Go, TypeScript, Git, Linux, SQLite, mDNS, TLS

- **Engineered a cross-platform background agent and CLI in Go** to capture local Git repository state, active branches, and uncommitted file diffs into portable JSON session schemas.
- **Designed a local network transfer layer using mDNS discovery and mutual TLS**, allowing zero-configuration device pairing to securely pipe workspace metadata between machines.
- **Developed TypeScript extensions utilizing the VS Code and Browser APIs** to extract active editor layouts, cursor positions, and tabs, achieving complete state restoration across Linux, macOS, and Windows.

GhostMirror — **Full-Stack AI Memory Engine** | React, TypeScript, TailwindCSS, FastAPI, Python, SQLite, Tauri

- Engineered a local-first desktop memory engine using **FastAPI**, **SQLite**, and **Tauri**, indexing **500+ developer events** with semantic retrieval for sub-100ms natural language queries.
- Developed a responsive **React/TypeScript** dashboard featuring semantic search, live event timelines, source filtering, and multi-theme support, reducing event retrieval latency by **~70%**.

Patient Voice Bot — **Real-Time AI Voice Agent Tester** | Python, FastAPI, OpenAI Realtime API, Twilio API, WebSockets

- Built a real-time **AI voice agent** integrating **Twilio Media Streams**, **OpenAI Realtime API**, and **WebSockets**, enabling low-latency conversational interactions while resolving transcript synchronization issues.
- Designed and validated **10+ prompt-driven patient scenarios** covering scheduling, insurance, and edge cases, capturing end-to-end call flows that streamlined QA testing and reduced manual validation effort by **~80%**.

LEADERSHIP:

AI, Coding, and Systems (ACS) Club — *Founder*

March 2026 – June 2026

- Founded a computer science organization promoting software engineering, **AI**, and **systems programming**, leading technical workshops and collaborative development sessions.
- Organized project-based learning initiatives and peer mentorship programs, fostering collaboration on full-stack applications, **Git workflows**, and **open-source development practices**.

PROFESSIONAL SKILLS:

Programming Languages: Python, JavaScript, C++, Go, Dart, TypeScript, SQL, PHP

Frameworks & Libraries: React, Flutter, FastAPI, Node.js, Express.js, Django, Tailwind CSS

Databases: SQLite, PostgreSQL, MySQL, MongoDB | **Developer Tools:** Postman, REST APIs, WebSockets, Tauri

Cloud & DevOps: Docker, GitHub Actions, CI/CD, Linux, Git